Face ladder v6: selection flow (images / people per step)

CelebA

CelebA (aligned crops)
202,599 images, 10,177 people
targets: every test-partition image (identities disjoint from train)
pool: 2,500 train identities with >= 3 images, 3 random images each (seed 2026)

Sampled
pool 7,500 img / 2,500 people
targets 19,962 img / 1,000 people

excluded: no single face found (MediaPipe Face Landmarker)
pool 81 img / 80 people
targets 157 img / 128 people

pool 7,419 img / 2,500 people
targets 19,805 img / 1,000 people

excluded: |yaw| or |pitch| >= 10 deg
pool 4,172 img / 2,211 people
targets 11,215 img / 980 people

pool 3,247 img / 1,938 people
targets 8,590 img / 978 people

excluded: identity inconsistent (ArcFace cosine to own centroid < 0.3)
pool 92 img / 46 people
targets 88 img / 58 people

pool 3,155 img / 1,892 people
targets 8,502 img / 973 people

excluded: face-skin mask < 2000 px
pool 0 img / 0 people
targets 1 img / 1 people

pool 3,155 img / 1,892 people
targets 8,501 img / 973 people

excluded: duplicate of another person's image (r > 0.9)
pool 18 img / 17 people
targets 33 img / 27 people

pool 3,137 img / 1,889 people
targets 8,468 img / 973 people

excluded: template fit failure (correlation < median - 3 MAD)
pool 97 img / 93 people
targets 188 img / 143 people

pool 3,040 img / 1,845 people
targets 8,280 img / 973 people

excluded: no usable repeat
(fewer than 2 usable images,
or only near-duplicates r > 0.9)
targets 50 people

Analysis set
923 targets (one most-frontal image per person with a repeat)
pool mean over 1,845 people (most-frontal image each)
expression strata: 457 neutral, 466 expressive

IMDb-WIKI

IMDb-WIKI crops
460,723 crops, 184,624 single-face (15,520 people)
targets: every single-face crop of every person in a Wikidata kinship pair
726 parent-child pairs, 601 sibling pairs
pool: 3,000 people outside kinship with >= 3 crops, 3 random crops each (seed 2026)

Sampled
pool 9,000 img / 3,000 people
targets 30,251 img / 1,327 people

excluded: no single face found (MediaPipe Face Landmarker)
pool 713 img / 636 people
targets 2,580 img / 665 people

pool 8,287 img / 2,996 people
targets 27,671 img / 1,296 people

excluded: |yaw| or |pitch| >= 10 deg
pool 5,521 img / 2,773 people
targets 18,867 img / 1,153 people

pool 2,766 img / 1,880 people
targets 8,804 img / 1,017 people

excluded: identity inconsistent (ArcFace cosine to own centroid < 0.3)
pool 136 img / 68 people
targets 434 img / 239 people

pool 2,630 img / 1,812 people
targets 8,370 img / 986 people

excluded: face-skin mask < 2000 px
pool 4 img / 4 people
targets 6 img / 6 people

pool 2,626 img / 1,810 people
targets 8,364 img / 986 people

excluded: duplicate of another person's image (r > 0.9)
pool 42 img / 39 people
targets 106 img / 84 people

pool 2,584 img / 1,780 people
targets 8,258 img / 971 people

excluded: template fit failure (correlation < median - 3 MAD)
pool 186 img / 181 people
targets 568 img / 270 people

pool 2,398 img / 1,668 people
targets 7,690 img / 949 people

excluded: no usable repeat
(fewer than 2 usable images,
or only near-duplicates r > 0.9)
targets 268 people

Analysis set
681 targets (one most-frontal image per person with a repeat)
pool mean over 1,668 people (most-frontal image each)
246 targets with a full sibling
expression strata: 381 neutral, 300 expressive

Pixel masks (not exclusions): face-skin class of MediaPipe Selfie Multiclass inside the face-oval hull and the photograph; illumination field division; 478-landmark affine to a symmetric template; z-score in the mask.
Amendment 7: glasses (template lens ellipses when triggered), facial hair, theatrical makeup and bright lipstick (CIELAB colour rules) removed pixel-wise; thresholds calibrated on CelebA attributes, see face_masks.